

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	17 June 2036
Team ID	
Project Name	VIP-C2-ShopEZ
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

S.No	Component	Description	Technology
1.	User Interface	Web UI for virtual trading, interactive dashboard, and live market explorer.	React, Vite, Bootstrap, Recharts (HTML, CSS, JavaScript)
2.	Application Logic-1	Backend API routing, secure JWT-based authentication, and user session management.	Node.js, Express.js
3.	Application Logic-2	Virtual transaction processing (Buy/Sell execution) and mock balance updates.	Node.js, Express.js
4.	Application Logic-3	Admin Control Room logic, role-based access control (RBAC), and platform-wide trade logging.	Node.js, Express Middleware
5.	Database	Storing user profiles, portfolios (holdings & \$100k balance), and transaction logs.	MongoDB (managed via MongoDB Compass)
6.	Cloud Database	Database Service on Cloud (Future scalability).	N/A for MVP (Utilizing local MongoDB)
7.	File Storage	Application source code, local environment configurations, and static assets.	Local Filesystem

8.	External API-1	Fetching real-time open/high/low stock prices for live market simulation.	External Cloud-based Stock API (e.g., Alpha Vantage, Finnhub)
9.	External API-2	Secondary integrations.	N/A for MVP
10.	Machine Learning Model	Predictive analytics or stock recommendations (Future scope).	N/A for MVP
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System. Local Server Configuration mapping frontend, backend, and DB.	Local System (Windows). Ports: 5173 (Frontend), 5000 (Backend), 27017 (Database)

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>